



```
In [1]: # importing major libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import plotly.express as px
```

```
In [2]: df = pd.read_csv('Employee_promotion.csv').iloc[:,0:5]
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Age	Salary	YearsExperience	NumCertifications	JobSatisfaction
0	18	138268	0	0	4
1	26	119117	5	1	6
2	21	105751	1	2	6
3	19	115759	0	0	6
4	31	120824	2	1	7

```
In [4]: df.describe()
```

```
Out[4]:
```

	Age	Salary	YearsExperience	NumCertifications	JobSatisfaction
count	1200.000000	1200.000000	1200.000000	1200.000000	1200.000000
mean	25.980833	130716.338333	4.527500	1.204167	5.983333
std	4.886583	31410.401415	3.112713	1.136615	1.054187
min	18.000000	57648.000000	0.000000	0.000000	1.000000
25%	22.000000	107870.750000	2.000000	0.000000	4.000000
50%	26.000000	126604.000000	4.000000	1.000000	6.000000
75%	29.000000	149426.000000	7.000000	2.000000	7.000000
max	43.000000	268467.000000	16.000000	6.000000	10.000000

```
In [5]: # plt.figure(figsize=(14,6))
# plt.plot(df.index,df.Age)
# plt.plot(df.index,df.Salary)
# plt.plot(df.index,df.YearsExperience)
# plt.plot(df.index,df.NumCertifications)
# plt.show()

px.line(df.head(150),title='Pattern got hidden due improper scaling')
```

```
In [6]: # standardisation
# classification
from sklearn.model_selection import train_test_split

x = df.iloc[:,0:-1]
y = df.JobSatisfaction

x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_stat
```

```
In [7]: print('rows for training',x_train.shape[0])
print('rows for testing',x_test.shape[0])
```

```
rows for training 960
rows for testing 240
```

```
In [8]: # standardisation

# algo --> fit (x_train)
# algo --> transform(x_train,y_train)
```

```
In [9]: from sklearn.preprocessing import StandardScaler
```

```
sc = StandardScaler()  
sc
```

Out[9]:

```
▼ StandardScaler ⓘ ?  
StandardScaler()
```

```
In [10]: sc.fit(x_train)
```

Out[10]:

```
▼ StandardScaler ⓘ ?  
StandardScaler()
```

```
In [11]: # sc.transform(x_train)
```

```
In [12]: x_train_sc = pd.DataFrame(sc.transform(x_train), columns = sc.get_feature_names_c  
x_train_sc.describe()
```

Out[12]:

	Age	Salary	YearsExperience	NumCertifications
count	9.600000e+02	9.600000e+02	9.600000e+02	9.600000e+02
mean	-5.551115e-17	2.979098e-16	-5.551115e-17	9.621933e-17
std	1.000521e+00	1.000521e+00	1.000521e+00	1.000521e+00
min	-1.619055e+00	-2.300981e+00	-1.455152e+00	-1.070611e+00
25%	-8.091058e-01	-7.303066e-01	-8.146449e-01	-1.070611e+00
50%	8.436974e-04	-1.191046e-01	-1.741379e-01	-1.761057e-01
75%	6.083058e-01	6.008751e-01	7.866227e-01	7.183995e-01
max	3.443129e+00	4.296377e+00	3.668904e+00	4.296420e+00

```
In [13]: x_test_sc = pd.DataFrame(sc.transform(x_test), columns = sc.get_feature_names_c  
x_test_sc.describe()
```

Out[13]:

	Age	Salary	YearsExperience	NumCertifications
count	240.000000	240.000000	240.000000	240.000000
mean	-0.015187	-0.071922	-0.026021	0.032612
std	0.945903	0.909002	0.983857	1.080824
min	-1.619055	-1.988260	-1.455152	-1.070611
25%	-0.606618	-0.720465	-0.814645	-1.070611
50%	0.000844	-0.244077	-0.174138	-0.176106
75%	0.608306	0.457377	0.466369	0.718400
max	3.443129	3.303670	3.348651	4.296420

```
In [14]: sc.get_feature_names_out()
```

```
Out[14]: array(['Age', 'Salary', 'YearsExperience', 'NumCertifications'],  
              dtype=object)
```

```
In [15]: # plt.figure(figsize=(14,6))  
# plt.plot(x_train_sc.index,x_train_sc.Age)  
# plt.plot(x_train_sc.index,x_train_sc.Salary)  
# plt.plot(x_train_sc.index,x_train_sc.YearsExperience)  
# plt.plot(x_train_sc.index,x_train.ScNumCertifications)  
# plt.show()
```

```
In [16]: import plotly.express as px  
px.line(x_test_sc.head(150),title='Standardisation')
```

```
In [17]: # normalisation
from sklearn.preprocessing import MinMaxScaler

norm = MinMaxScaler()
norm
```

```
Out[17]: ▼ MinMaxScaler ⓘ ?
MinMaxScaler()
```

```
In [18]: norm.fit(x_train)
```

```
Out[18]: ▼ MinMaxScaler ⓘ ?
MinMaxScaler()
```

```
In [19]: x_train_norm = pd.DataFrame(norm.transform(x_train), columns=norm.get_feature_r
x_train_norm.describe()
```

Out[19]:

	Age	Salary	YearsExperience	NumCertifications
count	960.000000	960.000000	960.000000	960.000000
mean	0.319833	0.348773	0.283984	0.199479
std	0.197646	0.151655	0.195260	0.186420
min	0.000000	0.000000	0.000000	0.000000
25%	0.160000	0.238076	0.125000	0.000000
50%	0.320000	0.330720	0.250000	0.166667
75%	0.440000	0.439851	0.437500	0.333333
max	1.000000	1.000000	1.000000	1.000000

```
In [20]: x_test_norm = pd.DataFrame(norm.transform(x_test),columns=norm.get_feature_names_out(),  
x_test_norm.describe())
```

Out[20]:

	Age	Salary	YearsExperience	NumCertifications
count	240.000000	240.000000	240.000000	240.000000
mean	0.316833	0.337871	0.278906	0.205556
std	0.186857	0.137783	0.192007	0.201382
min	0.000000	0.047401	0.000000	0.000000
25%	0.200000	0.239568	0.125000	0.000000
50%	0.320000	0.311777	0.250000	0.166667
75%	0.440000	0.418100	0.375000	0.333333
max	1.000000	0.849530	0.937500	1.000000

```
In [21]: px.line(x_test_norm.head(150),title='Normalisation')
```

